"Advanced Statistics for Physics Analysis"
Laboratory session 1 - report

Libero Pollini

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Overview

We are given a dataset of integer-valued observations (in an attached file):

$$D = \{x_1, x_2, ..., x_n\} \tag{1}$$

representing the number of events observed in identical time intervals. Our goal is to analyze the data using the Bayesian framework according to different assumptions on the likelihood and prior distributions. A first visualisation of the data is provided in figure 1.

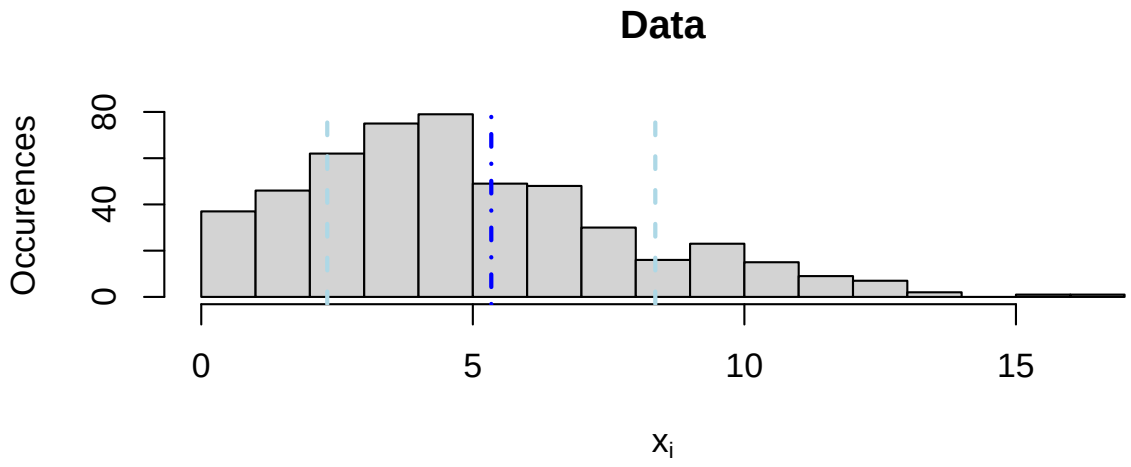


Figure 1: Histogram of the observed dataset ($n = 500$ integer-valued counts). The blue dashed vertical line marks the sample mean $x_{\text{mean}} \approx 5.34$; the light-blue dashed lines mark $x_{\text{mean}} \pm \text{STD}(x)$, where $\text{STD}(x) \approx 3.09$ is the sample standard deviation.

1 Poisson likelihood

We assume here that the data are generated from a Poisson distribution:

$$x_i \sim \text{Poisson}(\lambda) \equiv \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \tag{2}$$

and we use a Gamma prior on λ:

$$\lambda \sim \text{Gamma}(\alpha, \beta) \equiv \frac{\beta^{\alpha}}{\Gamma(\alpha)} \cdot \lambda^{\alpha-1} \cdot e^{-\beta\lambda} \tag{3}$$

1.a Derivation of the posterior distribution

Basing on these assumptions, the posterior distribution is simply (using the assumption of independent identically distributed samples, Bayes' theorem and the definitions of the Gamma and Poisson distributions, and doing some simple algebra):

$$p(\lambda|D) \propto p(D|\lambda)\cdot p(\lambda) = \prod_{i=1}^{n}\left(\frac{\lambda^{x_i}e^{-\lambda}}{x_i!}\right)\cdot\lambda^{\alpha-1}e^{-\beta\lambda} \propto \lambda^{\sum_i x_i+\alpha-1}\cdot e^{-(\beta+n)\lambda} \propto \text{Gamma}(\sum_i x_i+\alpha, \beta+n) \quad (4)$$

Upon naming $s := \sum_i x_i$, $\alpha' := s + \alpha$, and $\beta' = \beta + n$ we have therefore:

$$p(\lambda|D) = \text{Gamma}(\alpha', \beta') \quad (5)$$

And, being the Gamma distribution a standard one, we know its normalisation constant *a priori*. For our dataset, upon inspection, we have $s = 2670$ and $n = 500$ (see figures 1 and 2). Therefore, seeing the expressions of α' and β', we expect the choice of the (Gamma) prior parameters to have a negligible impact on the resulting posterior distribution. In our analysis, we choose $\alpha = 2$ and $\beta = 1$.

1.b MAP estimate of λ

To estimate the λ parameter through the "Maximum *a posteriori*" (MAP) method, i.e. by definition the parameter value corresponding to the maximum of the posterior distribution, we can simply look up the expression of such maximum for the Gamma distribution:

$$\lambda_{MAP} = \frac{\alpha' - 1}{\beta'} \approx 5.331337 \quad (6)$$

1.c Assessment of symmetry of posterior and estimate uncertainty

By plotting the posterior distribution (figure 2), one can already see that it is approximately symmetric (as expected: for n large, any posterior distribution tends to a Gaussian independently of the choice of the prior, as a consequence of the Bernstein - von Mises theorem). To quantify this symmetry, we use two simple ways:

- We check the difference between the mean and the mode (i.e. the MAP value) of the posterior distribution. The mean is simply $\lambda_{\text{mean}} = \frac{\alpha'}{\beta'} \approx 5.333333$. As the two values are very close, the (Gamma) posterior distribution is approximately symmetric.
- We evaluate the "skewness" of the distribution. For the Gamma distribution, this is simply $\frac{2}{\sqrt{\alpha'}} \approx 0.0387$: a value close to zero, leading to the same conclusion.

To evaluate the uncertainty on the estimated λ parameter, we can again use a simple known expression for the standard deviation Gamma distribution:

$$\text{STD}(\lambda) = \frac{\alpha'}{\beta'^2} \approx 0.103 \quad (7)$$

And write finally, thanks to this *de facto* symmetry:

$$\lambda = (5.33 \pm 0.10) \quad (8)$$

with a probability of about 68%. A comparison between data and the likelihood can be seen in figure 3.

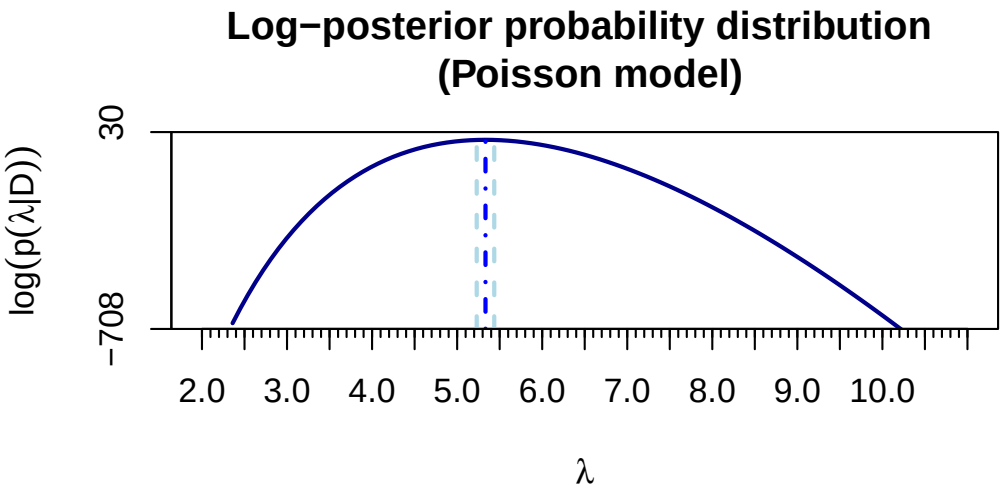


Figure 2: Log-posterior distribution $\log p(\lambda|D) = \text{Gamma}(\alpha', \beta')$ under the Poisson–Gamma model. The blue dashed vertical line marks the posterior mean $\lambda_{\text{mean}} \approx 5.33$; the light-blue dashed lines mark $\lambda_{\text{mean}} \pm \text{STD}(\lambda)$, with $\text{STD}(\lambda) \approx \pm 0.10$. The near-perfect symmetry on the log scale is consistent with the low skewness (≈ 0.039) of the Gamma posterior.

2 Posterior prediction distribution (same model)

2.a Derivation of the posterior prediction distribution

To derive the posterior predictive distribution for a new observation x_{new}, we simply use the law of total probability and evaluate the integral (over the support of the total distribution, Λ):

$$\begin{aligned} p(x = x_{\text{new}}|D) = \int_\Lambda d\lambda \cdot (x_{\text{new}}|\lambda) \cdot p(\lambda|D) = \int_0^{+\infty} d\lambda \cdot \frac{\lambda^{x_{\text{new}}} e^{-\lambda}}{x_{\text{new}}!} \cdot \frac{\beta'^{\alpha'}}{\Gamma(\alpha')} \lambda^{\alpha'-1} e^{-\beta'\lambda} = \\ = \frac{\beta'^{\alpha'}}{x_{\text{new}}! \cdot \Gamma(\alpha')} \int_0^{+\infty} d\lambda \cdot \lambda^{x_{\text{new}}+\alpha'-1} \cdot e^{-(\beta'+1)\lambda} = \\ = \frac{\beta'^{\alpha'}}{x_{\text{new}}! \cdot \Gamma(\alpha')} \cdot \frac{\Gamma(x_{\text{new}} + \alpha')}{(\beta'+1)^{x_{\text{new}}+\alpha'}} = \\ = \frac{\Gamma(\alpha' + x_{\text{new}})}{\Gamma(\alpha') \cdot x_{\text{new}}!} \cdot \left(\frac{\beta'}{1+\beta'}\right)^{\alpha'} \cdot \left(\frac{1}{1+\beta'}\right)^{x_{\text{new}}} = \\ = \binom{\alpha' + x_{\text{new}} - 1}{x_{\text{new}}} \cdot p_s^{\alpha'} \cdot (1-p_s)^{x_{\text{new}}} = \\ = \text{NegBin}(\alpha', p_s) \end{aligned} \tag{9}$$

Where in the fourth equality we simply recognised the normalisation of the Gamma distribution with parameters $\alpha'' := x_{\text{new}} + \alpha'$ and $\beta'' := \beta' + 1$; in the sixth one we recalled the property of the Gamma function, $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$, the definition of binomial coefficient, and performed the simple change of variable $p_s := \frac{\beta'}{1+\beta'}$; finally, in the last one, we recalled the definition of the Negative Binomial probability mass function. This distribution can be seen in figure 4.

2.b Cumulative posterior prediction distribution: an example

We now want to compute the probability that the next observation exceeds twice the empirical average of the dataset, $x_{\text{mean}} \approx 5.34$. All we have to do, in R, is to use the cumulative distribution function (CDF) of

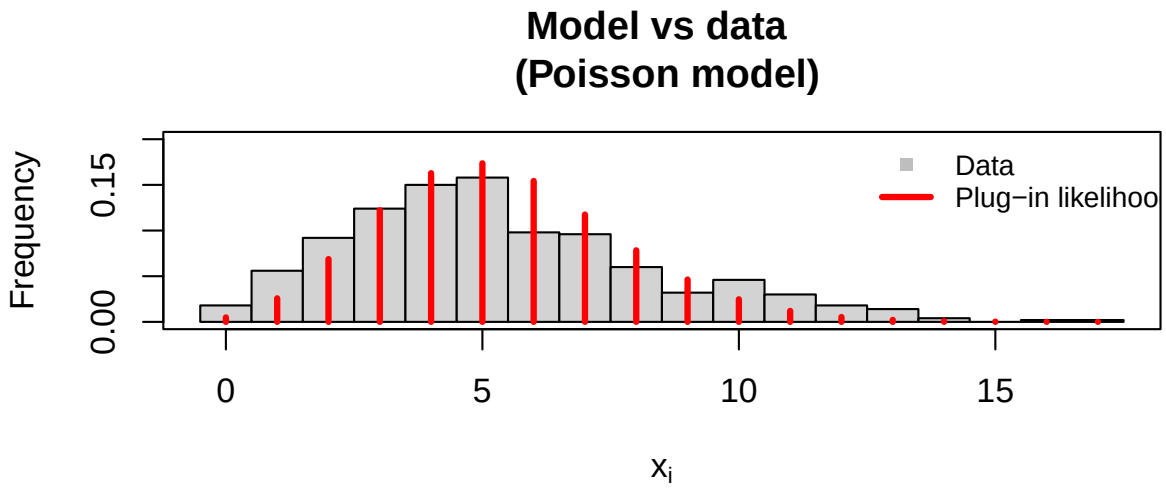


Figure 3: Comparison of the observed data (grey histogram, normalised to unit area) and the plug-in Poisson likelihood evaluated at the MAP estimate $\lambda_{\text{MAP}} \approx 5.33$ (red stems). The Poisson model provides a reasonable but imperfect fit, visibly underestimating the tails of the empirical distribution.

the negative binomial distribution (which has, at any rate, no simple analytical expression):

$$p(x_{\text{new}} > 2x_{\text{mean}}) = \sum_{x=2x_{\text{mean}}+1}^{+\infty} \text{NegBin}_x(\alpha', p_s) = 1 - \text{CDF}_{\text{NegBin}}(2 \cdot x_{\text{mean}}) \approx 2.09\% \tag{10}$$

3 Negative binomial likelihood

We consider now an alternative model with a Negative Binomial distribution for the ,

$$x_i \sim \text{NegBin}(r, \pi) \tag{11}$$

, where $r > 0$ and $0 < \pi < 1$ are the model parameters. For the whole dataset, therefore,

$$p(D|r, \pi) = \prod_{i=1}^{n} \left[\binom{x_i + r - 1}{x_i} \cdot (1-\pi)^{x_i} \cdot \pi^r \right] = \prod_{i=1}^{n} \left[\binom{x_i + r - 1}{x_i} \right] \cdot (1-\pi)^{\sum_i x_i} \cdot \pi^{nr} \tag{12}$$

3.a Choice of priors and posterior evaluation

We want to choose an appropriate prior and compute the corresponding posterior distribution. The choice of the prior actually corresponds to picking a prior for each of the two parameters, as the total prior will simply be the product of the two. The interval in which π is defined is the same as the support of the Beta distribution, which also happens to be the conjugate prior of the Negative Binomial distribution (from which we can expect a better analytical tractability); it is then natural to pick

$$\pi \sim \text{Beta}(\alpha_1, \beta_1) \equiv \frac{1}{\text{B}(\alpha_1, \beta_1)} \cdot (1-\pi)^{\beta_1 - 1} \cdot \pi^{\alpha_1 - 1} \tag{13}$$

where the normalisation constant is is the so-called "Beta function", defined as $\text{B}(z_1, z_2) \equiv \int_0^1 dt \cdot t^{z_1 - 1} \cdot (1-t)^{z_2 - 1}$. Let's pick now the prior distribution for the other parameter, r. We might be tempted to find a conjugate prior to the likelihood, which depends on r both through the π^r and through the binomial terms. However, there does not seem to exist such a distribution. Therefore, looking at the half-open interval in

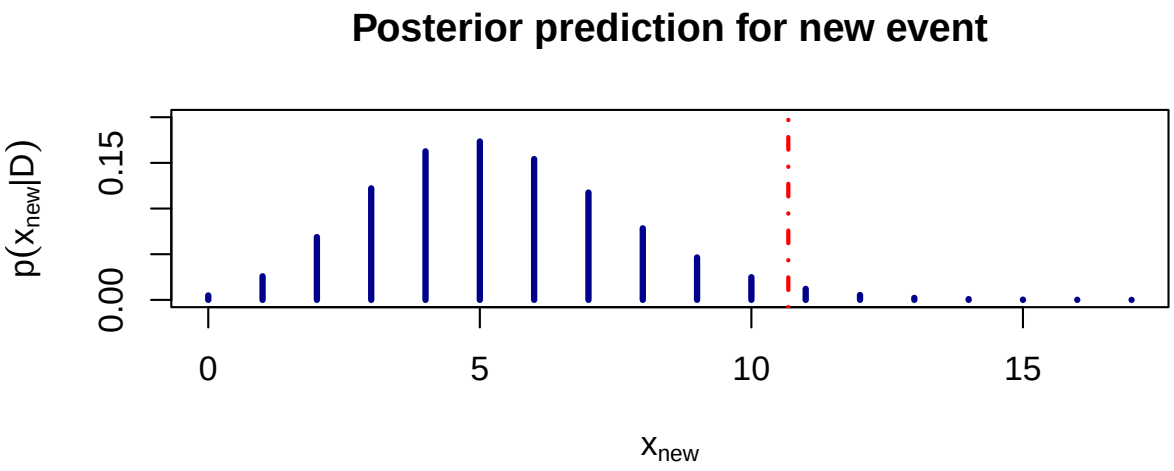


Figure 4: Posterior predictive distribution $p(x_{\text{new}}|D) = \text{NegBin}(\alpha', p_s)$ for a new observation (blue stems). The red dashed vertical line marks the threshold $2\bar{x} \approx 10.68$; the probability mass to its right equals $p(x_{\text{new}} > 2\bar{x}) \approx 2.09\%$.

which this parameter is defined, and keeping in mind that prior choice should be almost irrelevant for the final result, we choose as prior the Gamma distribution:

$$r \sim \text{Gamma}(\alpha_2, \beta_2) \equiv \frac{\beta_2^{\alpha_2}}{\Gamma(\alpha_2)} \cdot r^{\alpha_2 - 1} \cdot e^{-\beta_2 r} \tag{14}$$

As before, we do not expect the choice of parameters in the prior to be relevant; in our analysis we picked $(\alpha_1, \beta_1, \alpha_2, \beta_2) = (2, 2, 2, 1)$.
With this choice of priors, the posterior distribution is:

$$p(r, \pi|D) \propto p(D|r, \pi) \cdot p(r) \cdot p(\pi) \propto \prod_{i=1}^{n} \left[\binom{x_i + r - 1}{x_i} \right] \cdot (1 - \pi)^{\beta_1 - 1 + s} \cdot \pi^{\alpha_1 - 1 + nr} \cdot r^{\alpha_2 - 1} \cdot e^{-\beta_2 r} \tag{15}$$

The meaning of the proportionality symbol here is that the posterior distribution is equal to the above function in the variables (r, π), aside from the a normalisation constant. The value of this constant is of course the integral over the whole parameter space of this function. However, we cannot evaluate this definite integral analytically, and since the knowledge of the constant is not necessary in any part of the following analysis, we decided to simply omit this approximate calculation (which could be performed for example with Monte Carlo methods).

3.b MAP estimate of parameters and their uncertainty

Assuming a Gaussian shape (therefore symmetric and unimodal) of the posterior (as before, where we used a similar likelihood and saw that prior shape became irrelevant due to the relatively large n), we want to give a point estimate of the parameters r and π via MAP estimation and its associated error.
As we are dealing with the maximum of a very peaked distribution, it is standard practice to work with its logarithm (of course, the problem is equivalent because the logarithm is a monotonically increasing function) in order to give more stability to numerical approximations. Furthermore, if we then change the sign of the function, the problem becomes a minimisation one. For these reason we consider in the following the "negative log" posterior distribution instead of the original one.
One way to proceed would be to marginalise the joint posterior PDF over one parameter (but in this case

there is no closed form for $p(\pi|D)$, nor a very clean expression for $p(r|D)$), then minimise each with respect to the other parameter with some numerical method to find the maximum. For uncertainty, one would have to compute the 2x2 Fisher information matrix of the original (non-log) joint posterior distribution, $\mathrm{I}_{ij} \equiv -\partial_i \partial_j \log(p(D|r,\pi)) \quad i,j \in \{r,\pi\}$ (here the expectation operator appearing in the definition of the Fisher information is not included, as only a single dataset is available). This is, equivalently, the covariance matrix of the negative log-likelihood (2-variable) of the joint posterior. In a Gaussian (second order Taylor) approximation, in the maximum of the distribution this matrix is also approximately equal to the inverse (which can be calculated trivially in 2 dimensions) of the Hessian matrix of the negative log-likelihood. The uncertainty (standard deviation) on each of the two parameters is then the square root of the respective diagonal elements of this covariance matrix.
Keeping all of this into consideration, we can take a slightly different but equivalent path: we find the minimum of the joint negative log-posterior probability with a standard optimization method that is already implemented in R (the `optim(..., method ="L-BFGS-B", ...)` function), and gives us the Hessian matrix in that point "for free".
One crucial point here is that the normalisation constant (which we have not calculated) of the joint posterior distribution is irrelevant in all steps of this procedure: even when computing the second derivatives for the Hessian matrix, we are making use of the property of the derivative of a logarithm, $\frac{\partial}{\partial x}\log(kf(x)) = \frac{\partial}{\partial x}\log(f(x))$. In this sense, too, working with logarithms of probabilities or the probabilities themselves is completely equivalent in this kind of problems.
The results we obtained with this procedure are the following:

$$\begin{aligned} r &= (6.6 \pm 0.9) \\ \pi &= (0.55 \pm 0.03) \end{aligned} \tag{16}$$

The 2-dimensional, unnormalised log-posterior is shown in figure 5, from which we also inferred an initial guess for the argmaximum. The resulting (plug-in parameters) likelihood is shown in figure 6.

3.c Evidence of the Negative Binomial model

The "evidence" of a given model is, by the law of total probability (naming θ the generic, multi-dimensional parameter, Θ the support, and θ_{MAP} the argmaximum of the posterior distribution):

$$p(D|M_{NB}) = \int_{\Theta} d\theta \cdot p(D|M_{NB},\theta)\cdot p(\theta) \approx (2\pi)^{d/2}\cdot(\det(H^{-1}))^{1/2}\cdot p(D|\theta_{\mathrm{MAP}})\cdot p(\theta_{\mathrm{MAP}}) \tag{17}$$

Where the final result is a first order Laplace approximation of the integral (d is the sample dimension and H the Hessian matrix of the posterior calculated in θ_{MAP}), for an integrand that is approximately a normalised Gaussian (as is our case: the integrand is the normalised posterior distribution). We can apply this approximation to compute the evidence of the Negative Binomial model. Note that here we are using normalised probabilities, but again we did not have to compute the overall normalisation of the posterior because we already have the normalisation constants for both the likelihood and the prior. Computationally, we once again work by taking the logarithm of the leftmost and rightmost terms. We obtain the result:

$$\log(p(D|M_{NB})) \approx -1240 \tag{18}$$

3.d Evidence of the Poisson model and Bayesian comparison

For the Poisson model, we can compute the evidence directly:

$$\begin{aligned} p(D|M_{\mathrm{Poisson}}) = \int_0^{+\infty} d\lambda \cdot p(D|\lambda)\cdot p(\lambda) = \int_0^{+\infty} d\lambda \cdot \frac{\lambda^s}{\prod_i(x_i!)} e^{-n\lambda}\cdot\frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda} = \\ = \frac{1}{\prod_i(x_i!)}\frac{\beta^\alpha}{\Gamma(\alpha)}\int_0^{+\infty} d\lambda\cdot\lambda^{\alpha'}e^{-\beta'} = \frac{1}{\prod_i(x_i!)}\frac{\beta^\alpha}{\Gamma(\alpha)}\frac{\Gamma(\alpha')}{\beta'^{\alpha'}} \end{aligned} \tag{19}$$

Leading to the result:

$$\log(p(D|M_{Poisson})) \approx -1279 \tag{20}$$

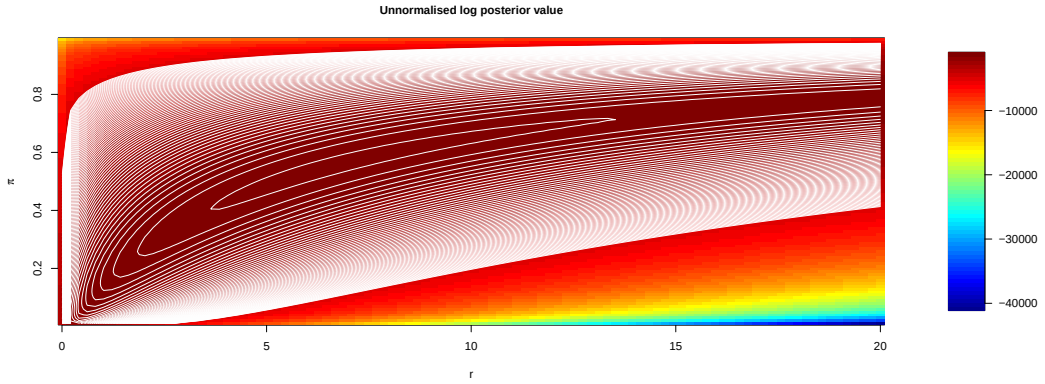


Figure 5: Unnormalised log-posterior $\log p(r, \pi|D)$ of the Negative Binomial model as a function of r and π, shown as a colour map with white iso-probability contours (spacing: 50 log-units). The sharp, well-localised maximum near $(r, \pi) \approx (6.6, 0.55)$ confirms the validity of the Gaussian (Laplace) approximation used for MAP estimation and uncertainty quantification.

We can compare this result with the evidence obtained for the Poisson model by computing the Bayes factor:

$$\mathrm{BF} = \frac{p(D|M_{\mathrm{NB}})}{p(D|M_{\mathrm{NB}})} = \exp\{\log(p(D|M_{NB})) - \log(p(D|M_{Poisson}))\} \approx 5.7 \cdot 10^{16} \tag{21}$$

As BF $>>$ 1, the Negative Binomial model is extremely favored with respect to the Poisson model. Seeing how the posterior distributions fit the data (figures 3 and 6 respectively), this is perhaps unsurprising: the Negative Binomial distribution seems to account for the "fat" tails of the sample. However, a fact that might be puzzling at first sight is that both evidences are extremely low: in other words, the probability of obtaining the given dataset, assuming each sample follows the given distribution, is very low in any case. This is actually a common occurrence for large dataset: the probability of obtaining a *specific* dataset is low because there is a huge ensemble of similarly "likely" dataset from which we are randomly sampling. In conclusion, one should always keep in mind that the Bayesian evidence of a given model has no intrinsic meaning, but must always be compared to that of other competing models to decide between them.

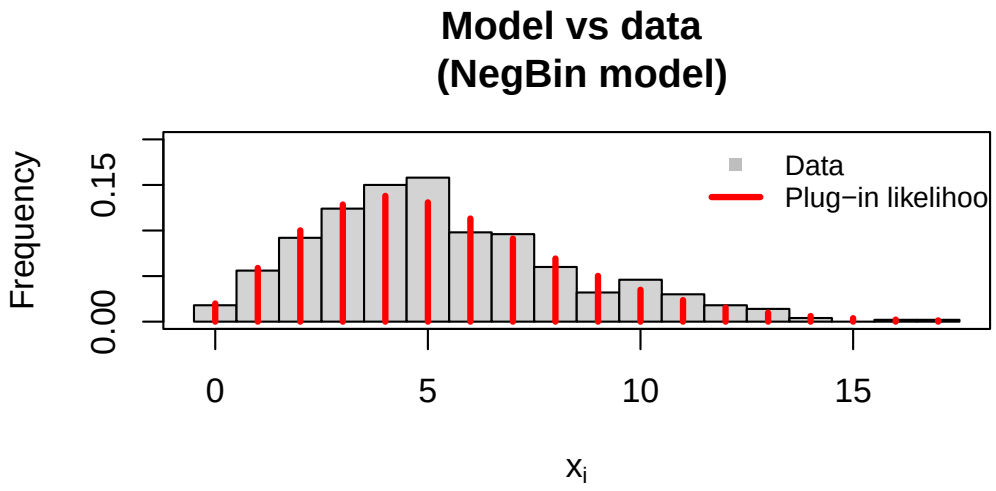


Figure 6: Comparison of the observed data (grey histogram, normalised to unit area) and the plug-in Negative Binomial likelihood evaluated at the MAP estimates $r_{\text{MAP}} \approx 6.6$, $\pi_{\text{MAP}} \approx 0.55$ (red stems). The NegBin model captures the overdispersion of the data significantly better than the Poisson model (see Figure 3), consistent with the overwhelming Bayes factor in its favour.

Code

The complete R code (for all sections) follows, commented where necessary. Output, where needed, is also included as comments.

```r
library(fields)

# import dataset
data <- scan("C:/Users/Libero/uni/1B/adv_stats/assignment_1/dataset.txt", what=
    integer())

# explore it
typeof(data)
#[1] "integer"
length(data)
#[1] 500
x_mean <- mean(data)
cat(x_mean)
#5.34
x_std <- sd(data)
cat(x_std)
#3.019708

s <- sum(data)
cat(s)
#2670
n <- length(data)
cat(n)
#500

# data histogram
data_histo <- hist(data, breaks=20, main="Data", xlab= expression(x[i]), ylab="
```

```
    Occurences") # sqrt(500) = 22,...
ymax <- max(data_histo$counts)
abline(v = x_mean, col = "blue", lty = 4, lwd=2)
abline(v = x_mean + c(-1, 1) * x_std, col = "light blue", lty = 2, lwd = 2)

############################################################
#                     POISSON MODEL
############################################################

# pick parameters (ininfluent)
alpha=2
beta=1

# change of variables
alpha_prime <- s + alpha
beta_prime <- beta + n
cat(alpha_prime, beta_prime)
#2672 501

# estimates from posterior
lambda_map=(alpha_prime-1)/beta_prime
posterior_mean=alpha_prime/beta_prime
posterior_std=sqrt(alpha_prime / beta_prime **2)
skewness=2.0/sqrt(alpha_prime)
cat(lambda_map, posterior_mean, posterior_std, skewness)
#5.331337 5.333333 0.1031764 0.03869116

# log posterior plot
curve(dgamma(x, shape = alpha_prime, rate = beta_prime),
      from = 2, to = 11, log = "y", axes = FALSE,
      main = "Log-posterior probability distribution \n (Poisson model)",
      xlab = expression(lambda), ylab = expression(log(p(lambda*"|"*D))), col = "
          dark blue", lwd = 2)
axis(1, at = seq(2, 11, by = 0.1), labels = FALSE, tcl = -0.2)
axis(1, at = seq(2, 11, by = 0.5), labels = TRUE)
usr <- par("usr")
y_lo <- 10^usr[3]
y_hi <- 10^usr[4]
axis(2, at = c(y_lo, y_hi), labels = format(log(c(y_lo, y_hi)), digits = 2,
    scientific = FALSE))
abline(v = posterior_mean, col = "blue", lty = 4, lwd = 2)
abline(v = posterior_mean + c(-1, 1) * posterior_std, col = "light blue", lty = 2,
     lwd = 2)
box()

# plug-in likelihood vs data plot (Poisson model)
x_vals <- 0:max(data)
hist(data, prob = TRUE, breaks = seq(min(data)-0.5, max(data)+0.5, by = 1),
     main = "Model vs data \n (Poisson model)", xlab = expression(x[i]), ylab="
         Frequency", ylim=c(0,0.2))
points(x_vals, dpois(x_vals, lambda=lambda_map), col = "red", type = 'h', lwd=3)
legend("topright", legend = c("Data", "Plug-in likelihood"),
       col = c("gray", "red"), pch = c(15, NA), lty = c(NA, 1),
       lwd = c(NA, 3), pt.cex = 0.8, cex = 0.8, bty = "n")
box()

p_s <- beta_prime/(1+beta_prime)

# posterior prediction plot
```

```r
plot(0, type = "n", xlim = range(x_vals), ylim = c(0, 0.2),
     xlab = expression(x[new]), ylab = expression(p(x[new]*"|"*D)), main = "
         Posterior prediction for new event")
points(x_vals, dnbinom(x_vals, size=alpha_prime, prob=p_s), col = "dark blue",
    type = 'h', lwd=3)
abline(v = 2*x_mean, col = "red", lty = 4, lwd=2)

# p(x>2*x_mean) calculation
cdf_2_x_mean = pnbinom(2*x_mean, size=alpha_prime, prob=p_s)
prob_excess= 1- cdf_2_x_mean
cat(prob_excess)
#0.02092887

############################################################
#                     NEGBIN MODEL
############################################################

# pick parameters
alpha_1=2
beta_1=2
alpha_2=2
beta_2=1

# log of the unnormalised posterior - function definition
log_unnorm_posterior <- function(r, pi) {
  log_bin <- sum(lchoose(data + r - 1, data)) # binomial coefficient part
  log_rest <- (alpha_1 - 1 + n * r) * log(pi) +
              (beta_1 - 1 + s) * log(1 - pi) +
              (alpha_2 - 1) * log(r) - beta_2 * r
  return(log_bin + log_rest)
}

# 2-var joint posterior plot
# grid of r and pi values
r_vals <- seq(0.01, 20, length.out = 100)
pi_vals <- seq(0.01, 0.99, length.out = 100)
grid <- expand.grid(r = r_vals, pi = pi_vals)
grid$log_z <- mapply(log_unnorm_posterior, grid$r, grid$pi)
# reshape to matrix for image/contour
z_mat <- matrix(grid$log_z, nrow = length(r_vals), ncol = length(pi_vals))
image.plot(r_vals, pi_vals, z_mat, xlab = "r", ylab = "  ",
           main = "Unnormalised log posterior value")
contour(r_vals, pi_vals, z_mat, add = TRUE, col = "white", # white contours
        levels = seq(-5000, 0, by = 50), drawlabels = FALSE)


# minimisation of negative log-likelihood
neg_log_unn_posterior <- function(params) -log_unnorm_posterior(params[1], params
    [2])
result <- optim(par=c(7.5,0.5), fn=neg_log_unn_posterior, method="L-BFGS-B",
    hessian=TRUE,
                lower=c(0.01,0.01), upper=c(100,0.99))
# because only method for box costraints multi-dim functions; in pi=0 or 1 log not
     defined
result
#$par
#[1] 6.6155459 0.5533274
#
#$value
```

```r
#[1] 1238.389
#
#$counts
#function gradient
#      14       14
#
#$convergence
#[1] 0
#
#$message
#[1] "CONVERGENCE: REL_REDUCTION_OF_F <= FACTR*EPSMCH"
#
#$hessian
#           [,1]       [,2]
#[1,]   35.00149  -903.6251
#[2,] -903.62510 24194.5119

# inspect result of minimisation
r_map <- result$par[1]
pi_map <- result$par[2]
covariance_matrix <- solve(result$hessian)
delta_r <- sqrt(covariance_matrix[1,1])
delta_pi <- sqrt(covariance_matrix[2,2])
cat(delta_r)
#0.8935039
cat(delta_pi)
#0.03398453

# log evidence of NegBin model - calculation
log_lik_map <- sum(dnbinom(data, size=r_map, prob=pi_map, log=TRUE))
log_prior_map <- dgamma(r_map, alpha_2, beta_2, log=TRUE) +
                 dbeta(pi_map, alpha_1, beta_1, log=TRUE)

log_evidence <- log_lik_map + log_prior_map + log(2*base::pi) + 0.5*log(det(
    covariance_matrix))
cat(log_evidence)
#-1239.919

# log evidence of Poisson model - calculation
log_evidence_pois <- -sum(lgamma(data+1)) +
                      alpha*log(beta) - lgamma(alpha) +
                      lgamma(alpha_prime) - alpha_prime*log(beta_prime)
cat(log_evidence_pois)
#-1278.505
bf=exp(log_evidence-log_evidence_pois)
cat(bf)
#5.7222e+16

# plug-in likelihood vs data plot (NegBin model)
hist(data, prob = TRUE, breaks = seq(min(data)-0.5, max(data)+0.5, by = 1),
     main = "Model vs data \n (NegBin model)", xlab = expression(x[i]), ylab="
         Frequency", ylim=c(0,0.2))
points(x_vals, dnbinom(x_vals, size=r_map, prob=pi_map),
       col="red", type='h', lwd=3)
legend("topright", legend = c("Data", "Plug-in likelihood"),
col = c("gray", "red"), pch = c(15, NA), lty = c(NA, 1),
lwd = c(NA, 3), pt.cex = 0.8, cex = 0.8, bty = "n")
box()
```